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Persistence of a biocontrol *Pseudomonas* inoculant as high populations of culturable and non-culturable cells in 200-cm-deep soil profiles

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A B S T R A C T

Little is known about the ecology of soil inoculants used for pathogen biocontrol, biofertilization and bioremediation under field conditions. We investigated the persistence and the physiological states of soil-inoculated *Pseudomonas protegens* (previously *Pseudomonas fluorescens*) CHA0 (10^8 CFU g^{-1} surface soil) in different soil microbial habitats in a planted ley (*Medicago sativa* L.) and an uncovered field plot. At 72 days, colony counts of the inoculant were low in surface soil (uncovered plot) and earthworm guts (ley plot), whereas soil above the plow pan (uncovered plot), and the rhizosphere and worm burrows present until 1.2 m depth (ley plot) were survival hot spots (10^5–10^6 CFU g^{-1} soil). Interestingly, strain CHA0 was also detected in the subsoil of both plots, at 10^2–10^5 CFU g^{-1} soil between 1.8 and 2 m depth. However, non-cultured CHA0 cells were also evidenced based on immunofluorescence microscopy. Kogure's direct viable counts of nutrient-responsive cells showed that many more CHA0 cells were in a viable but non-culturable (VBNC) or a non-responsive (dormant) state than in a culturable state, and the proportion of cells in those non-cultured states depended on soil microbial habitat. At the most, cells in a VBNC state amounted to 34% (above the plow pan) and those in a dormant state to 89% (in bulk soil between 0.6 and 2 m) of all CHA0 cells. The results indicate that field-released *Pseudomonas* inoculants may persist at high cell numbers, even in deeper soil layers, and display a combination of different physiological states whose prevalence fluctuates according to soil microbial habitats.

1. Introduction

The effective use of bacterial inoculants for biocontrol, biofertilization or bioremediation requires their release at large cell numbers into the soil ecosystem. Information on the persistence, dissemination and activity of the introduced bacteria is crucial for the evaluation of their efficacy and ecological impact in soil (Keel and Défago, 1997; Jäderlund et al., 2008). Often, the population of introduced rhizobacteria declines gradually after field release, due to abiotic stress and negative interactions from resident microbiota (Ryder et al., 1994; Weller and Thomashow, 1994; Keel and Défago, 1997; Fischer et al., 2010). However, field soil is heterogeneous and is comprised of contrasted microbial habitats, whose potential for inoculant survival may differ.

Introduced plant-beneficial rhizobacteria may successfully colonize the rhizosphere, because they are well adapted to this environment (Weller and Thomashow, 1994; Keel and Défago, 1997; Fliessbach et al., 2009; Fischer et al., 2010). However, after application of these rhizobacteria as a soil drench or seed treatment, a considerable number of the cells may be transported from sources of inoculum to non-target sites (e.g. bulk soil in deeper layer) by water flow or by other vectors such as earthworms (Doube et al., 1994; Natsch et al., 1996; Langenbach et al., 2006; Nechitaylo et al., 2010). Soil conditions encountered by introduced rhizobacteria at non-target sites may differ considerably from those in the rhizosphere in terms of water activity, nutrient availability, temperature, and resident microbial communities (Weller and Thomashow, 1994; Richter and Markewitz, 1995; Fischer et al.,

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2010). However, whereas rhizosphere colonization by released bacteria is amply documented, little is known about inoculant behavior at non-target sites outside the rhizosphere. For an adequate assessment of the ecology of microbial inoculants, therefore, more information is needed on their fate at soil sites other than those where they are expected to display their beneficial activity.

To date, monitoring of microbial inoculants has relied largely upon colony counts (Ryder et al., 1994; Fliessbach et al., 2009; Fischer et al., 2010), but stressed cells from otherwise culturable species that are present in a non-cultured state are not accounted for (Pedersen and Leser, 1992; Binnerup et al., 1993; Chapon et al., 2003; Arana et al., 2010; Buck and Oliver, 2010). Quantitative PCR is a promising alternative (Chapon et al., 2003; von Felten et al., 2010) but may not be reliable either when bacteria are subjected to stress (Rezzonico et al., 2003) or dead cells are also present (Keer and Birch, 2003). The ecological significance of non-cultured bacterial cells in the environment remains debated. They may represent a transient state towards cell death in some cases, but in others cells may persist extensively in this state, raising the possibility that it could represent a mechanism for bacterial persistence under adverse environmental conditions (Binnerup and Sørensen, 1993; Kragelund and Nybroe, 1994; Troxler et al., 1997b; Oliver, 2010). Indeed, there is evidence that non-cultured cells might return to a state of active growth in certain cases (Grey and Steck, 2001; Du et al., 2007; Gorshkov et al., 2009; Kana and Mizrahi, 2010).

One approach to characterize the physiological state of inoculant cells is to monitor them using a combination of colony counts, direct microscopic counts of cells marked with immunofluorescence (IF) staining (Pedersen and Leser, 1992; Heijnen et al., 1995; Hase et al., 2000; Mascher et al., 2003), and direct viable counts (DVC) of nutrient-responsive cells with the Kogure method (Kogure et al., 1979; Troxler et al., 1997b; Mascher et al., 2002). In the latter, viable/substrate responsive cells become enlarged following treatment with yeast extract and nalidixic acid, whereas dormant or dead cells do not respond to the treatment and remain small. The DVC procedure has been applied successfully in the soil environment (Heijnen et al., 1995; Hase et al., 2000; Mascher et al., 2003).

In the current work, the fate of the model inoculant *Pseudomonas protegens* (previously *Pseudomonas fluorescens*; Ramette et al., 2011) CHA0 in target and non-target soil locations of 2-m-deep field soil profiles was investigated at 72 days after release. Strain CHA0 is a well-characterized, root-colonizing bacterium that produces several antifungal compounds and protects plants from a range of soilborne diseases (Keel and Défago, 1997; de Werra et al., 2011). It has also been considered for insect control (Péchy-Tarr et al., 2008) and *n*-alkane biodegradation (Smits et al., 2002). In a previous study at the same field site, heavy rainfall led to rapid transport of high numbers of CHA0 cells to deeper soil layers, i.e. to at least 150-cm depth, through paths of preferential water flow (Natsch et al., 1996). The objectives of this follow-up study were (i) to investigate the spread and survival/persistence of strain CHA0 in surface soil and in deeper soil layers, several months after release and (ii) to compare the occurrence and frequency of different physiological states (i.e. culturable, viable but non-culturable, and dormant states) of the strain in various soil microbial habitats.

2. Materials and methods

2.1. Inoculum preparation

The experiment was done with strain CHA0-Rif (Natsch et al., 1994), a spontaneous rifampin-resistant mutant of *P. protegens* strain CHA0 (Stutz et al., 1986) that is equivalent to the wild-type in terms of grow *in vitro*, stress resistance and root-colonization ability (Natsch et al., 1994). *P. protegens* CHA0-Rif was cultivated on King's medium B (KB) agar (King et al., 1954) containing 100 μg rifampin per ml. For inoculation to soil, cells were harvested after 16 h of growth on KB agar at 27 °C and resuspended in distilled water. The suspension was filtered through three layers of cheesecloth and adjusted to 10^{10} CFU per mL.

2.2. Soil and climatic conditions at the field site

P. protegens CHA0-Rif was released at the end of May, over a 140 cm × 140 cm area in each of two neighboring field plots, one whose soil surface was exposed and the other whose surface was covered with a well-established ley (*Medicago sativa* L.). The two inoculation areas were 10 m apart. Soil structure was better in the ley plot as it was not tilled for four years and displayed higher root and earthworm densities (Natsch et al., 1996). The ley was cut to a height of 5 cm to enable homogeneous application of strain CHA0-Rif. The uncovered plot was tilled each year and had a plow pan at about 20-cm depth. Wheat (*Triticum aestivum* L.) shoots (at flowering stage) were cut-off and completely removed prior to bacterial release, leaving root systems in situ. Both plots correspond to the same clayey loam (gleyic Cambisol; FAO soil group), which is a typical type of soil found in the Swiss midlands. Soil material is of alluvial origin, with a coarse gravel layer (below 80–100 cm) overlaid with clayey and loamy layers. Each of the two inoculation areas was about 5 m from those used for monitoring vertical dissemination of strain CHA0-Rif in a previous study (Natsch et al., 1996). Climatic conditions, soil temperature and relative soil water content at the field site during the experiment are presented in Fig. S1. Total rainfall over the duration of the experiment was 182 mm (including an artificial rain of 40 mm applied shortly after inoculation; see below).

2.3. Application of bacteria and chemical tracers

Strain CHA0-Rif and chemical tracers were applied to the field plots as described by Natsch et al. (1996). Briefly, 1 L of a bacterial suspension (10^{10} CFU of CHA0-Rif per mL) was applied per m^2 of surface soil in May, using a Birchmeier backpack compression sprayer (Birchmeier, Stetten, Switzerland). Then, a heavy rainfall was simulated by application of 40 mm of an irrigation solution over a period of 8 h with an automatic sprinkler (Flury et al., 1994; Natsch et al., 1996). The irrigation solution contained 0.015 mol of potassium bromide (KBr) per L (i.e. 0.8 mol m^{-2}) as a mobile, conservative tracer and 0.005 mol of the food coloring Brilliant Blue FCF (Plüss and Stauffer, Basel, Switzerland) per L to identify the flow paths of percolating water (Flury et al., 1994; Natsch et al., 1996).

2.4. Preparation of soil profiles and definition of microbial habitats

In both plots, a 210-cm-deep trench was dug 72 days after field application of CHA0-Rif. Three vertical soil profiles (100-cm wide and 200-cm deep) were then prepared at right angles from each original trench, and at a distance of 20 cm from each other, using sterilized knives, as described by Natsch et al. (1996).

Microbial habitats were defined as follows. Rhizosphere soil samples consisted of roots and closely-associated soil; they were found throughout the first 60 cm of the ley plot, where soil was clayey to loamy (Natsch et al., 1996). There was no rhizosphere soil per se in the uncovered plot, as wheat shoots had been removed and root system remains were considered as part of bulk soil. Samples from earthworm burrows correspond to material (mostly earthworm casts) lining the burrows, which extended from the soil surface to about 120-cm depth in the ley plot. Earthworms were not prevalent enough in the uncovered plot to be considered in the study. Plow pan samples refer to the thin (about 5-cm thick) soil layer in the uncovered plot that was located immediately above the plow pan, at a depth of about 20 cm. There was no plow pan in the ley plot, which had not been tilled. Bulk soil samples were taken in the uncovered plot, away from root remains and at all depths (except that soil immediately above the plow pan was considered separately), and in the uncovered plot, away from worm burrows and below 60 cm (i.e. below the root zone).

2.5. Extraction and preparation of samples

Soil samples were extracted at different depths of each soil profile (150 samples per plot) with sterilized knives or by pressing sterilized steel cores (internal diameter 2 cm) 1.5 cm deep into the soil. Control samples were taken at different depths in the non-inoculated part of the soil profiles (10 samples per plot). For each sample, the location within the profile and the corresponding microbial habitat (rhizosphere, worm burrow, plow pan, bulk soil) were recorded. Each soil sample (0.4–4 g) was transferred to a 25-mL flask containing 18 mL sterile distilled water, shaken for 1 h at 300 rpm, and vortexed for 15 s. The resulting soil suspension was immediately used for enumeration of bacteria.

In addition, earthworms were collected in worm burrows in the ley plot. The earthworms were killed by a 5-s immersion in water at 48 °C and immediately dissected with a sterilized scalpel (Parle, 1963). For each earthworm, the whole gut (0.25 g ± 0.15 g) was transferred to 1.5 mL sterile distilled water with sterilized forceps. The samples were kept on ice (at the most 30 min) prior to 15-s vortexing and bacterial enumeration.

2.6. Enumeration of culturable bacteria and measurement of tracer concentration

For enumeration of culturable cells of CHA0-Rif, vortexed sample suspensions were decimally diluted and plated on KB agar containing rifampin (100 μg mL^{-1}) and actidione antifungal (188 μg mL^{-1}). No rifampin-resistant bacteria were found in the control samples. When randomly-chosen rifampin-resistant colonies were studied by randomly amplified polymorphic DNA analysis, as described (Keel et al., 1996), results confirmed that they corresponded to CHA0-Rif.

Plating was also done on 1/10 strength tryptic soy agar (TSA, Difco) and *Pseudomonas* medium S1 (Gould et al., 1985) to count total culturable heterotrophic bacteria and total culturable fluorescent pseudomonads, respectively. Bromide concentration was measured directly in the soil suspensions with an ion-selective electrode (Ingold Messtechnik, Urdorf, Switzerland). The detection limit was 1.5×10^{-8} mol Br^- per gram of soil.

2.7. Direct viable counts (DVC) and total cell counts of CHA0-Rif cells

For DVC counts (Kogure et al., 1979) of strain CHA0-Rif, soil suspension aliquots containing 20 mg soil (i.e. 0.1–1 mL) were added to 10 mL sterile distilled water. After vortexing for 1 min, 0.025% (w/v) yeast extract and 0.002% (w/v) nalidixic acid were added and the suspensions were incubated at room temperature (20–22 °C) in the dark for 6 h. The samples were then fixed with formaldehyde (final concentration 2% v/v) and examined by IF microscopy, as described below. Control samples were not treated with yeast extract and nalidixic acid but were fixed immediately with formaldehyde.

Indirect IF staining was used for detection of CHA0-Rif cells, based on a strain-specific polyclonal antiserum (Troxler et al., 1997a) used with fluorescein isothiocyanate-conjugated anti-rabbit immunoglobulins, as described (Mascher et al., 2003). Observations were made with a Zeiss Axioskop epifluorescence microscope (filters 450–490 nm, objective 40× and/or 100×). Counting was carried out over the entire filter (at least 50 microscopic fields and/or 200 bacterial cells) and two filters per sample were examined. Only soil samples containing more than 10^5 CHA0-Rif cells g^{-1} soil (i.e. the IF detection limit) were examined. CHA0-Rif cells that were responsive to the DVC treatment elongated to a length of 4–10 μm (rods with flagella) and were counted as viable/responsive cells, whereas non-responsive cells were spherical with diameter of 0.8–1.2 μm (no visible flagella but cell wall apparently intact) and were considered to be dormant. No cross-reaction with other soil bacteria could be observed in soil samples from areas of the field plots where no CHA0-Rif cells had been introduced.

The number of CHA0-Rif cells in a VBNC state was computed by subtracting colony counts from DVC counts of the strain. The number of CHA0-Rif cells in a dormant state was computed by subtracting DVC counts from total IF counts of the strain.

2.8. Statistics

Cell numbers were log-transformed and percentages were arcsine-transformed before computing means and standard deviations, and performing ANOVA and Fisher's LSD tests. Statistics were carried out at $P < 0.05$, using SYSTAT for Windows (SYSTAT, Chicago, IL). In the vertical soil profiles, means shown in the figures were calculated by considering all individual samples within each 10-cm-deep soil layer from the three profiles per plot. Pearson coefficient ($P < 0.05$) was used for correlation analysis between mobile tracer concentration and number of culturable CHA0-Rif cells.

3. Results

3.1. Culturable CHA0-Rif cells in vertical soil profiles of two field plots

At 72 days after release, culturable cells of CHA0-Rif in the 200-cm-deep vertical soil profiles were present in 93% of the 163 samples from the ley plot and 97% of the 145 samples from the uncovered plot (detection limit 50 CFU g^{-1} soil). In the top 3 cm, the number of culturable cells of CHA0-Rif remained relatively high (often about 10^5 CFU g^{-1} soil) in the ley plot (Fig. 1A) but was near or below detection limit in the uncovered plot (Fig. 1B). In the ley plot, the numbers of CHA0-Rif cells typically ranged between 10^4 and 5×10^5 CFU g^{-1} soil between 3 and 150 cm deep, but declined to about 10^2 CFU g^{-1} at further depths (Fig. 1A). In the uncovered plot, in contrast, culturable counts were high for the soil immediately above the plow pan (at 15–20 cm depth), whereas they decreased sharply immediately below (to about 10^2 CFU g^{-1} soil), then reached about 10^5 CFU g^{-1} soil at a 100-cm depth and remained at this level till 200 cm depth (Fig. 1B).

3.2. Relation between bromide tracer and culturable CHA0-Rif cells in soil profiles

At 72 days, the mobile chemical tracer potassium bromide reached the highest concentrations (about 10^{-6} mol g^{-1} soil) in the surface soil of both plots (Fig. 2). In the soil layers below, the patterns of vertical distribution of bromide in the two soil profiles were not the same at 20 cm, at 60–80 cm and below 160 cm depth, indicating some (moderate) variation in the flow paths of percolating water in the two profiles. There was no correlation (Pearson coefficient; $P < 0.05$) between mobile tracer concentration and number of culturable CHA0-Rif cells. In addition, bromide was not

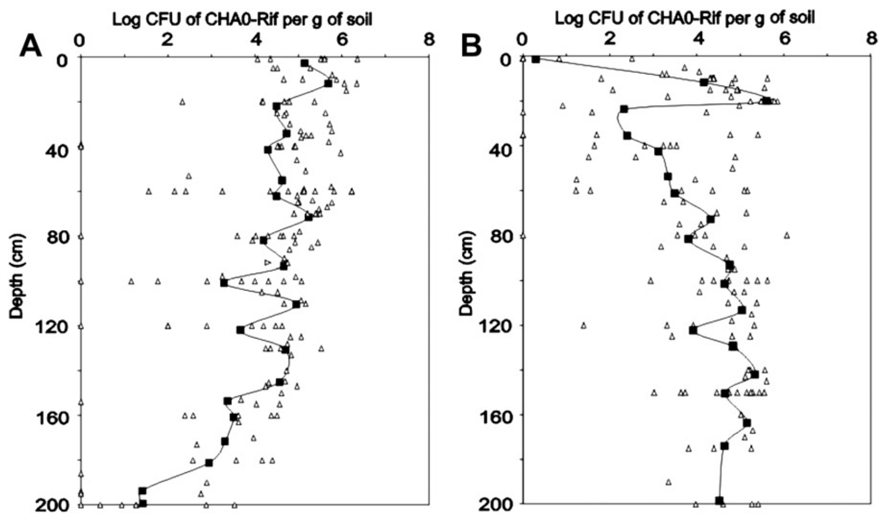

Fig. 1. Number of culturable cells of *Pseudomonas* sp. CHA0-Rif at different depths in vertical soil profiles of a ley (A) and an uncovered (B) field plot. Soil samples ($n = 150$) from three profiles per plot were analyzed 72 days after application of 10^{13} CFU of strain CHA0-Rif per m^2 of soil surface. Δ = Single values from individual soil samples; ■ = means calculated from all log-transformed values per 10-cm-deep soil layer. In adjacent field plots, the mean population density of CHA0-Rif one day after release was 5–6.2 log CFU per gram of soil between 20 cm and 140 cm depth (36).

found in 11% of all soil samples (detection limit of 1.5×10^{-8} mol g^{-1} soil) but culturable CHA0-Rif cells were evidenced in 90% of these samples (and when so, at $>2 \times 10^5$ CFU g^{-1} of soil in 45% of them). In samples where bromide was found, the ratio between culturable CHA0-Rif cell number and bromide concentration fluctuated greatly, i.e. from 4.5×10^6 to 1.5×10^{13} CFU mol^{-1} bromide (average 1.0×10^{12} CFU mol^{-1} bromide).

3.3. Culturable CHA0-Rif cells in different soil habitats

At 72 days, the rhizosphere and the soil material lining earthworm burrows (ley plot), as well as the moist soil layer immediately above the plow pan (uncovered plot) contained more than 10^5 CFU g^{-1} soil of strain CHA0-Rif (Table 1). This is also where colony counts of resident culturable heterotrophic bacteria reached 4×10^7 CFU g^{-1} soil or more. Colony counts of CHA0-Rif were stable at about 2×10^5 CFU g^{-1} from the soil surface to a depth of 60 cm, both for rhizosphere soil and worm burrows, but they dropped to 5×10^4 CFU g^{-1} for worm burrows between 60 and 100 cm (not shown). In bulk soil (both plots), colony counts of CHA0-Rif were lower and displayed higher standard deviations, indicating a heterogeneous distribution of the cells.

In the ley plot, the blue tracer dye (which identifies the flow path of percolating water) was visually detected in the fresh gut material of all earthworms, and since all blue-colored (rhizosphere) soil samples contained more than 5×10^5 CFU of CHA0-Rif g^{-1} soil, it means that earthworms had fed on soil containing strain CHA0-Rif. However, culturable CHA0-Rif cells were below the detection limit (5×10^2 CFU g^{-1} fresh gut material) in 27 of the 30

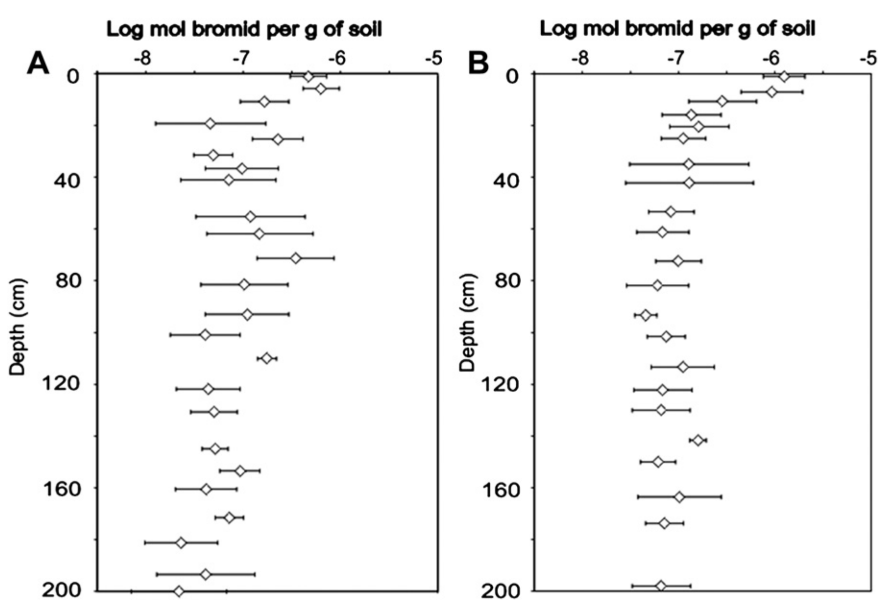

Fig. 2. Concentration of the chemical tracer bromide at different depths in vertical soil profiles of a ley (A) and an uncovered (B) field plot, at 72 days after application. Means $\pm$ standard deviations were calculated from all log-transformed values per 10-cm-deep soil layers from three soil profiles per plot.

Table 1
Number of culturable cells (log CFU per g; means ± standard deviations) of *Pseudomonas protegens* CHA0-Rif and resident bacteria in different soil microbial habitats in the first 200-cm soil depth of a ley plot and an uncovered field plot 72 days after inoculation.

Microbial habitat	*P. protegens* CHA0-Rif	Heterotrophic bacteria	Fluorescent pseudomonads
Ley plot			
Rhizosphere soil (0–60 cm; $n = 40$[a])	5.3 ±± 0.7	7.8 ± 0.4	5.9 ± 0.5
Bulk soil (60–200 cm; $n = 30$)	3.5 ± 1.6	6.8 ± 0.6	5.2 ± 0.1
Worm burrows (0–120 cm; $n = 30$)	5.2 ± 0.6	7.6 ± 0.5	Not done
Earthworm gut (0–120 cm; $n = 30$)	Not applicable[b]	7.2 ± 0.7	4.3 ± 0.7
Uncovered plot			
Bulk soil (0–200 cm; $n = 30$)	4.6 ± 0.9	6.4 ± 0.6	5.1 ± 0.1
Above plow pan (15–20 cm; $n = 25$)	5.2 ± 0.7	7.8 ± 0.5	Not done

[a] Number of samples.
[b] In 27 of 30 earthworm guts the number of culturable cells of CHA0-Rif was below the detection limit (which was 2.7–3.0 log CFU per gram of gut), and in the three others it was 3.4, 4.0 and 4.3 log CFU per gram of gut.

earthworms examined, even though indigenous fluorescent pseudomonads were present at 2×10^4 CFU g^{-1} (versus at least 5×10^5 g^{-1} in the surrounding soil) (Table 1).

3.4. Physiological states of CHA0-Rif cells in different soil microbial habitats

A total of 60 soil samples from either plot for which total cell counts exceeded the IF detection limit of 10^5 CHA0-Rif cells g^{-1} soil was examined to probe the physiological state of inoculant cells. This was done for various soil microbial habitats except in earthworm guts, where no CHA0-Rif cell was found by IF microscopy. Cell culturability of CHA0-Rif cells was more than 90% in the surface soil of both field plots at one day after release (our unpublished data), but by 72 days it had dropped to 25–32% in worm burrows and the rhizosphere, and even to 4–5% above the plow pan and in other bulk soil samples (Table 2).

CHA0-Rif cells in a VBNC state were evidenced in all soil microbial habitats. They did not represent more than 10% of all inoculants cells, except in humid soil immediately above the plow pan in the uncovered plot, where they reached 34% (Table 2). However, the majority of CHA0-Rif cells were in a dormant state, even in the rhizosphere, and they were up to 89% of all CHA0-Rif cells in bulk soil.

Table 2
Cells of *Pseudomonas protegens* CHA0-Rif in distinct physiological states (expressed as percent of the total number of CHA0-Rif cells applied) according to soil microbial habitats in field plots 72 days after inoculant release.

Microbial habitat	Culturable state	VBNC state	Dormant state
Rhizosphere soil ($n = 7$)[a]	32 ± 11[b] f[c]	10 ± 15 g	58 ± 11 e
Bulk soil[d] ($n = 12$)	4.0 ± 3.4 f	7.0 ± 18 f	89 ± 26 e
Worm burrows ($n = 7$)	25 ± 20 f	8.0 ± 8.0 f	67 ± 22 e
Above plow pan ($n = 3$)	5.0 ± 5.0 e	34 ± 33 e	61 ± 45 e

[a] Number of samples.
[b] Standard deviation.
[c] Statistical comparisons between the three physiological states (within rows) are shown using letters e.g. ($P < 0.05$).
[d] Data from the ley and uncovered plots were pooled as they did not differ at $P < 0.05$.

3.5. Detection and distribution of CHA0-Rif cells in the soil profiles

The estimation of CHA0-Rif cell numbers provides indication on the amount of introduced CHA0-Rif cells that moved to, survived/died or perhaps even multiplied in different soil layers. This estimation was made on the (simplistic) assumption that randomly-chosen samples were representative of the soil layer they originated from. At 72 days, it was estimated that the number of CHA0-Rif cells in the 200-cm soil profiles represented 30% (ley plot) and 23% (uncovered plot) of the number of inoculant cells applied (Table 3). Two thirds of CHA0-Rif cells detected were above 130 cm in the ley plot but below 130 cm in the uncovered plot.

When physiological states were considered, culturable CHA0-Rif cells represented only 3.2% (ley plot; most of them above 60 cm, i.e. where roots were present) and 1.0% (uncovered plot) of the total number of applied CHA0-Rif cells (Table 3). VBNC cells were detected mostly at 3–130 cm in the ley plot, versus at 3–60 cm and at 130–200 cm in the uncovered plot. Dormant cells were found in majority at 60–130 cm in the ley plot and at 130–200 cm in the uncovered plot.

4. Discussion

The current experiment was set up to monitor field establishment and spread of the model inoculant *P. protegens* CHA0-Rif in various microbial habitats of soil profiles. Field monitoring of bacterial inoculants has been assessed on many occasions (e.g. Troxler et al., 1997b; Jäderlund et al., 2008; Gamalero et al., 2009), but often without much consideration for the heterogeneity of soil vertical profiles and of inoculant physiological status.

In this field experiment, culturable CHA0-Rif cells were found mostly along the walls of macropores one day after release (Natsch et al., 1996). By 72 days, however, the inoculant cells had spread from the macropores and colonized soil micropores (detection in 95% of the soil samples). Comparable behavior has been reported with other bacterial inoculants (Recorbet et al., 1995), but here the findings were extended to field conditions and, strikingly, to deeper soil depth (2 m). Significant rainfall was simulated immediately after release and occurred naturally afterward (Fig. S1A), leading to preferential flow and (except in the first few cm in the exposed field) high soil water content (Fig. S1B,C,D). This was likely to

Table 3
Distribution of cells of *Pseudomonas protegens* CHA0-Rif in different physiological states in vertical soil profiles 72 days after inoculant application to a ley and an uncovered field plot.

Application and detection	Total number of cells		% of the total number of applied cells		
	Cells m^{-2}	%	Culturable state	VBNC state	Dormant state
Ley plot					
Applied initially	1.1×10^{13}	100	90	10	<0.1
Detected 0–3 cm	7.2×10^{10}	0.7	0.2	0.1	0.4
Detected 3–60 cm	7.2×10^{11}	6.6	2.1	0.7	3.8
Detected 60–130 cm	2.2×10^{12}	19.8	0.8	1.4	17.6
Detected 130–200 cm	3.5×10^{11}	3.1	0.1	0.2	2.8
Total detected	3.3×10^{12}	30.2	3.2	2.4	24.6
Uncovered plot					
Applied initially	1.1×10^{13}	100	90	10	<0.1
Detected 0–3 cm	2.2×10^{6}	<0.001	<0.001	BD[a]	BD
Detected 3–60 cm	4.4×10^{11}	4.0	0.2[b]	1.4	2.4
Detected 60–130 cm	5.3×10^{11}	4.8	0.2	0.3	4.2
Detected 130–200 cm	1.6×10^{12}	14.1	0.6	1.0	12.5
Total detected	2.6×10^{12}	22.9	1.0	2.7	19.1

[a] BD, below detection limit.
[b] Most culturable CHA0-Rif cells between 3 and 60 cm were found in the 5-cm-deep soil layer immediately above the plow pan.

promote bacterial dissemination (Hagedorn et al., 1978; Natsch et al., 1996). However, the number of culturable CHA0-Rif cells did not correlate with bromide tracer concentration. This may have entailed (i) retention of inoculant cells in certain micropores by straining in the soil matrix (Tan et al., 1992), (ii) active bacterial motility (Reynolds et al., 1989), as 45% of samples without detectable amount of bromide contained in excess of 2×10^5 CFU CHA0-Rif g^{-1}, and/or (iii) higher protozoan grazing in macropores (Görres et al., 1999).

As anticipated, the number of culturable CHA0-Rif cells depended also on soil microbial habitat. It ranged from very low (first few cm of the uncovered plot, which was poorly rooted and exposed to drying) to high (e.g. rhizosphere soil of the ley plot) (Fig. 1). High survival on roots was expected for a rhizobacterium (Thompson et al., 1992; Jäderlund et al., 2008).

In the uncovered plot, the greatest number of CHA0-Rif cells was just above the plow pan (Fig. 1), which tends to be water-logged (including here at sampling). It is likely that the plow pan reduced further vertical spread of the inoculants cells transported to this layer. Certain bacteria (e.g. *Ralstonia solanacearum*; Akiew and Trevorrow, 1994) and fungi (*Fusarium oxysporum*; Burke, 1968) are known to accumulate above plow pans, but so far this was mainly documented for phytopathogens.

Earthworms can promote microbial dispersal in soil including for pseudomonads (Doube et al., 1994). However, culturable cells of CHA0-Rif were hardly present in earthworm gut samples, in agreement with previous findings on other bacterial inoculants (Heijnen and Marinissen, 1995). Indeed, indigenous culturable pseudomonads displayed lower counts within earthworms in comparison with bulk soil, unlike the total culturable heterotrophic bacteria (Table 1). In contrast, a large number of culturable CHA0-Rif cells were found in the material (mostly earthworm cast) lining the burrows (Table 2), probably because the few CHA0-Rif cells that survived gut passage proliferated in this organic-rich material. We conclude that earthworms were not a major vector for direct distribution of the inoculant in soil yet could promote it indirectly by providing survival hot spots (i.e. burrows).

In our experiment, a large fraction of the introduced cells were non-cultured at 72 days (more than 90% based on estimates; Table 3), and the latter included a combination of cells in VBNC state and (predominantly) cells in dormant state. This observation is in accordance with previous findings on *Enterobacter cloacae* (Pedersen and Leser, 1992), *Rhizobium leguminosarum* (Postma et al., 1988), *Helicobacter pylori* (Buck and Oliver, 2010) and pseudomonads (Binnerup et al., 1993; Chapon et al., 2003) including *P. protegens* CHA0-Rif (Troxler et al., 1997b; Hase et al., 2000; Mascher et al., 2003). Here, this issue was taken one step further by demonstrating that the percentage of non-culturable cells in true field conditions depended on the type of soil microbial habitat defined over the 2-m-deep soil profiles. To date, cell culturability of *P. protegens* CHA0-Rif in the field was only assessed in topsoil (Troxler et al., 1997b).

The occurrence of non-culturable cells of strain CHA0-Rif in bulk soil was not surprising, as inoculant cells may be exposed to oligotrophy (Williams, 1985) and other stress factors (van Elsas and van Overbeek, 1993), which may promote loss in cell culturability (Hase et al., 1999; Mascher et al., 2000). The largest fraction of CHA0-Rif cells in a VBNC state was detected in the soil just above the plow pan. Water-logging, non-decomposed plant material and visual signs of iron reduction indicated anaerobic conditions. Oxygen deficiency in *Pseudomonas aeruginosa* resulted in cells that were non-culturable when studied under aerobic conditions (Binnerup and Sørensen, 1993), and similar findings were made *in vitro* with *P. protegens* CHA0-Rif (Mascher et al., 2000). Furthermore, when plow pan conditions (i.e. oxygen limitation combined with reducing conditions) were simulated in soil microcosms, large amounts of CHA0-Rif cells were found in a VBNC state (Mascher et al., 2003). Therefore, the current study validates the findings of Mascher et al. (2003) by demonstrating their relevance under agronomic field conditions. Non-culturable CHA0-Rif cells tend to occur as small spheres (Troxler et al., 1997b), which was confirmed here when examining samples in which non-cultured cells were very predominant. Since smaller cells have better percolation abilities than large cells (Fontes et al., 1991; Gannon et al., 1991), it could be that small non-cultured cells were transported preferentially during the experiment, which might have contributed to the higher percentages of non-cultured CHA0-Rif cells in bulk soil above the plow pan and in deeper soil layers.

The inoculant displayed high percentage of culturable cells in the rhizosphere and worm burrows, but non-culturable CHA0-Rif cells occurred as well in these microbial habitats (Table 2). Troxler et al. (1997b) and Hase et al. (2000) reported this for the rhizosphere, but not for earthworm burrows. These results mean that ecological conditions there were not favorable enough to maintain cell culturability of the inoculant, perhaps because of abiotic stress (Troxler et al., 1997b).

In conclusion, this work demonstrated that as many as 10^{12} cells of *P. protegens* CHA0-Rif were detectable in soil at 72 days after application of 10^{13} inoculant cells, with field population levels as high as 10^5 CFU g^{-1} soil at 2 m depth. However, strain persistence fluctuated according to soil microbial habitat (with survival high spots in the rhizosphere, earthworm burrows, and soil above plow pan), and a majority of inoculants cells were not detectable by standard cultivation technique (i.e. in a VBNC or especially a dormant state). Understanding the ecological role of released bacteria that have switched to a non-cultured state will be important to predict their impact in the environment.

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Appendix. Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.soilbio.2011.09.020.

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